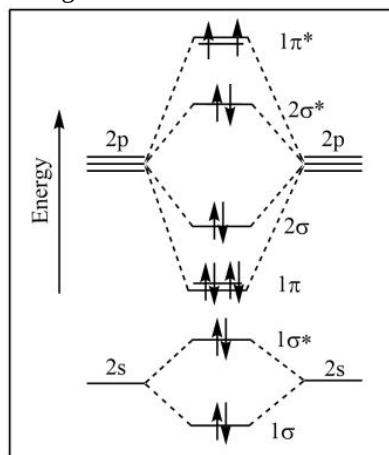
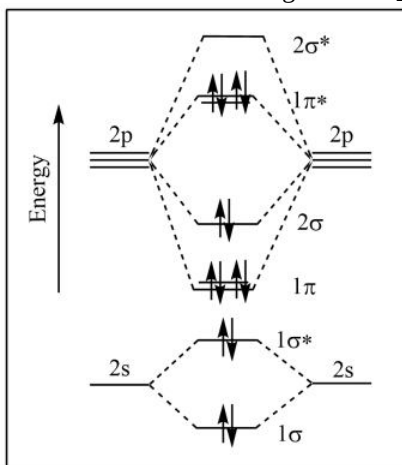


CHEMISTRY

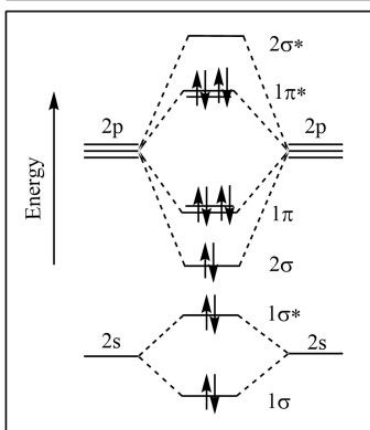
SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

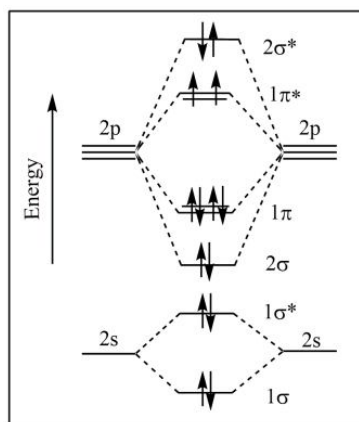
*Q. 1 The correct molecular orbital diagram for F_2 molecule in the ground state is
(A)



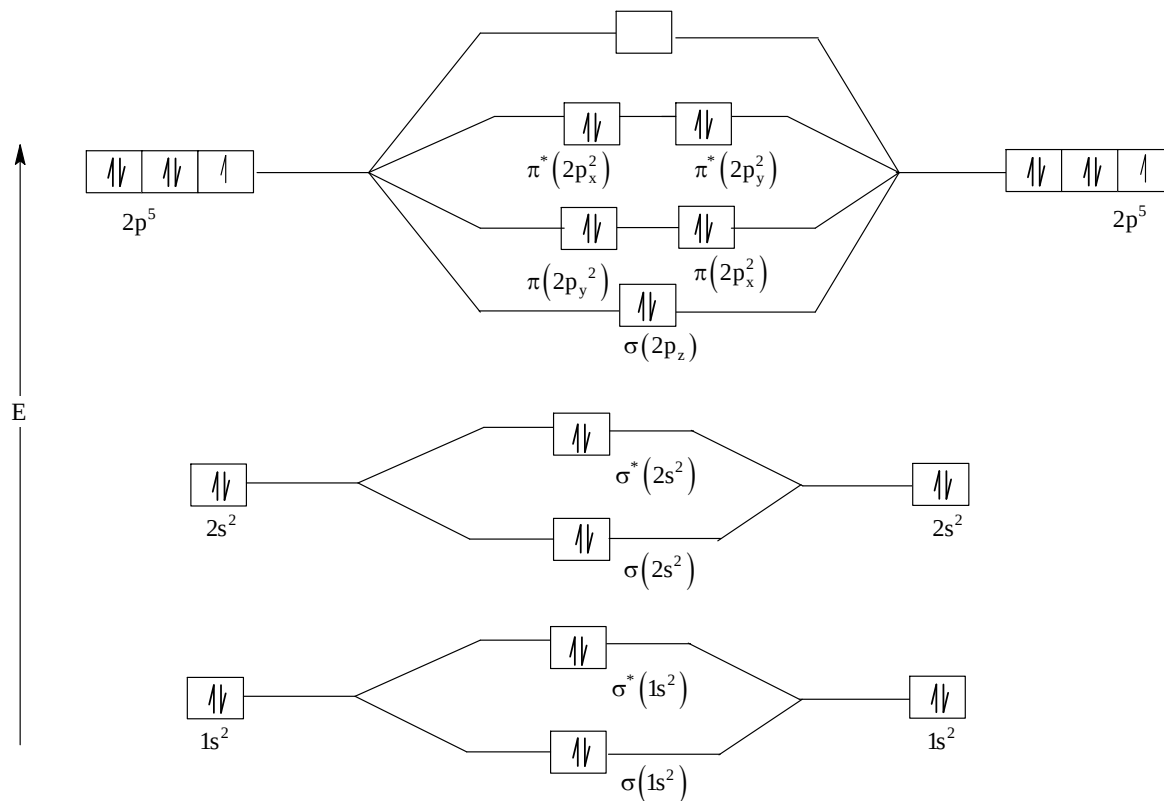
(C)



(D)



Sol. C



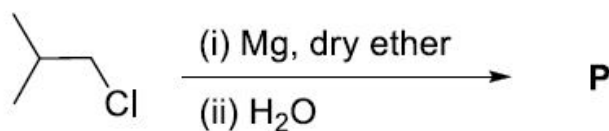
- Q. 2 Consider the following statements related to colloids.
- (I) Lyophobic colloids are **not** formed by simple mixing of dispersed phase and dispersion medium.
 - (II) For emulsions, both the dispersed phase and the dispersion medium are liquid.
 - (III) Micelles are produced by dissolving a surfactant in any solvent at any temperature.
 - (IV) Tyndall effect can be observed from a colloidal solution with dispersed phase having the same refractive index as that of the dispersion medium.

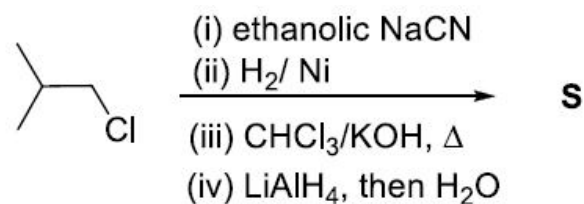
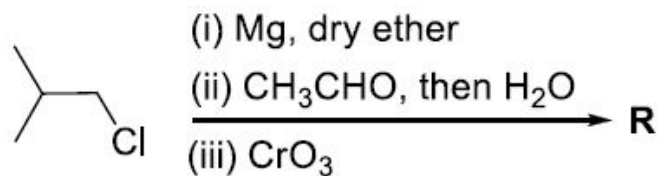
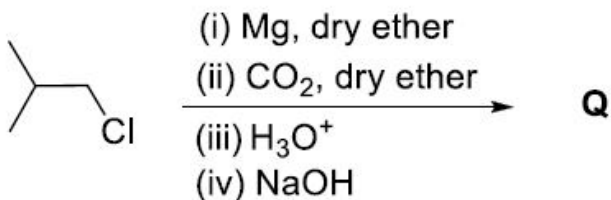
The option with the correct set of statements is

- | | |
|--------------------|--------------------|
| (A) (I) and (II) | (B) (II) and (III) |
| (C) (III) and (IV) | (D) (II) and (IV) |

- Sol. **A**
- In tyndall effect, refractive indices of dispersed phase and dispersion medium differ greatly in magnitude.
- Micelles are formed by surfactant at CMC or above CMC and at Kraft temperature or above Kraft temperature.

- Q. 3 In the following reactions, **P**, **Q**, **R**, and **S** are the major products.

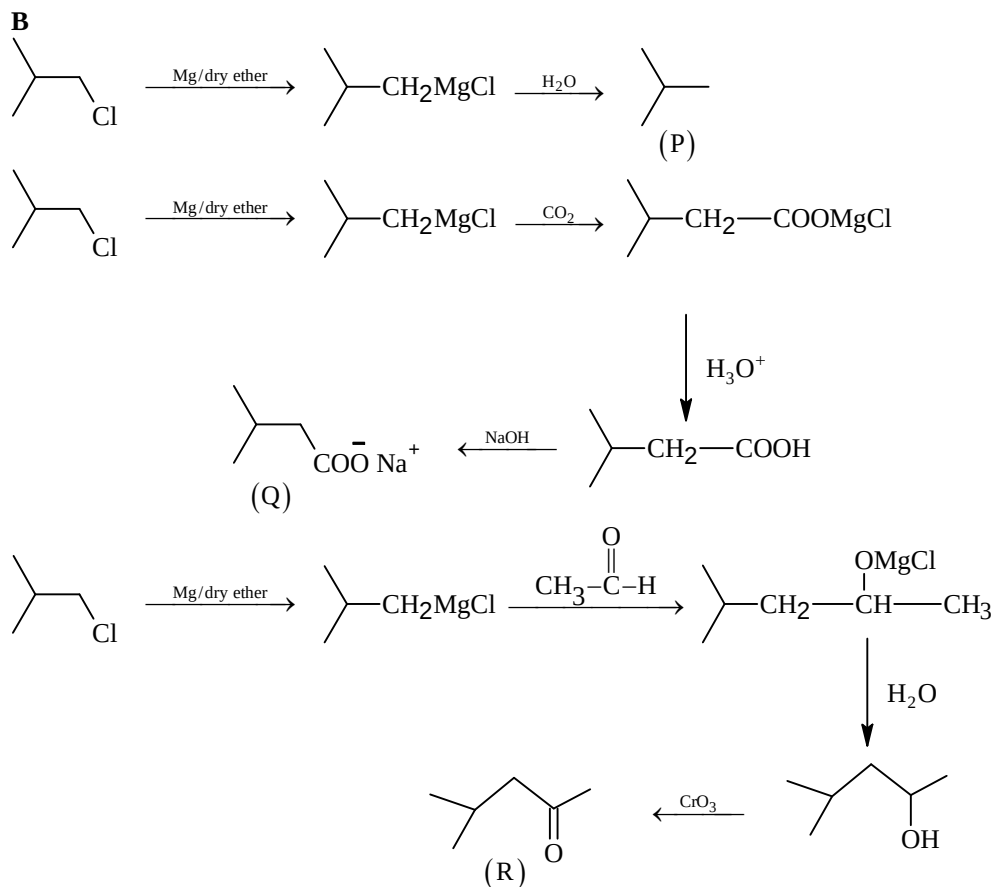


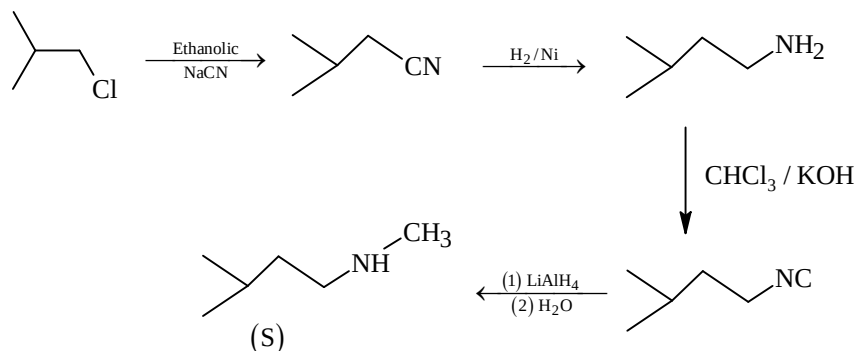


The correct statement about **P**, **Q**, **R**, and **S** is

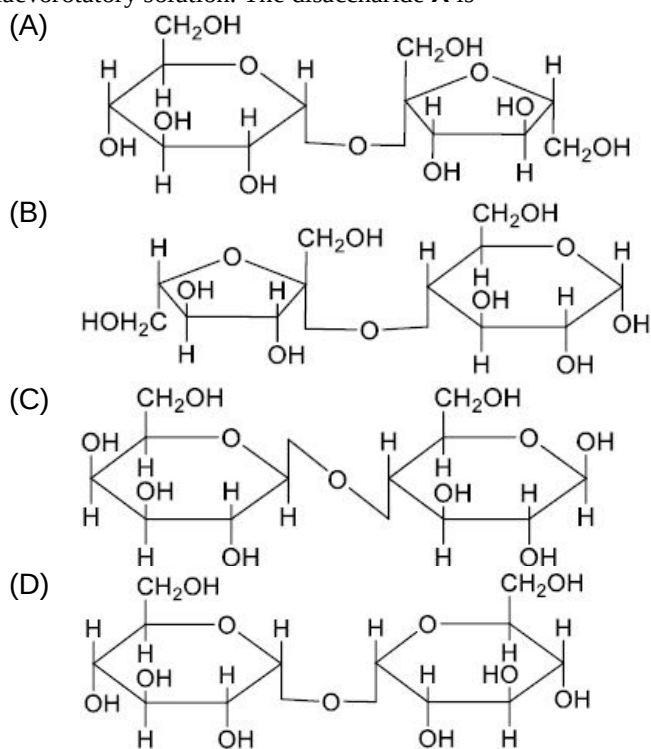
- (A) **P** is a primary alcohol with four carbons.
 (B) **Q** undergoes Kolbe's electrolysis to give an eight-carbon product.
 (C) **R** has six carbons and it undergoes Cannizzaro reaction.
 (D) **S** is a primary amine with six carbons.

Sol.

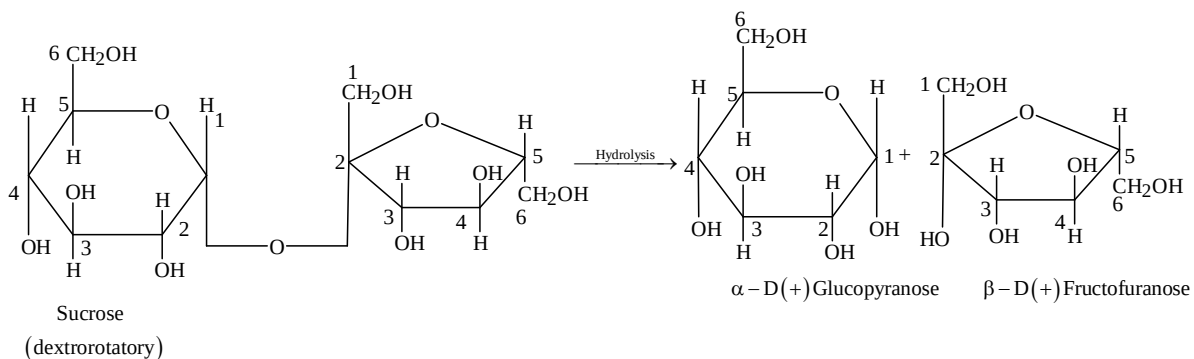




Q.4 A disaccharide **X** cannot be oxidised by bromine water. The acid hydrolysis of **X** leads to a laevorotatory solution. The disaccharide **X** is



Sol. A



Hydrolysis of sucrose brings about a change in the sign of rotation from dextro(+) to laevo(-) and the product named as invert sugar.

SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
- Negative Marks* : -2 In all other cases. · For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 mark;
 choosing **ONLY** (B) will get +1 mark;
 choosing **ONLY** (D) will get +1 mark;
 choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 choosing any other option(s) will get -2 marks.

- Q.5 The complex(es), which can exhibit the type of isomerism shown by $[\text{Pt}(\text{NH}_3)_2\text{Br}_2]$, is(are)
 $[\text{en} = \text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2]$
- (A) $[\text{Pt}(\text{en})(\text{SCN})_2]$ (B) $[\text{Zn}(\text{NH}_3)_2\text{Cl}_2]$
 (C) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ (D) $[\text{Cr}(\text{en})_2(\text{H}_2\text{O})(\text{SO}_4)]^+$

Sol. C, D

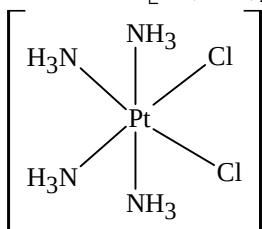
$[\text{Pt}(\text{NH}_3)_2\text{Br}_2]$ is a square planar complex.

The given compound can show geometrical isomerism (cis-trans form)

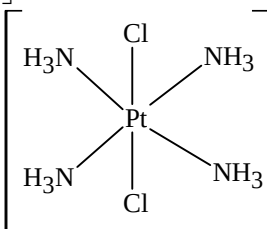
Option (A) : $[\text{Pt}(\text{en})(\text{SCN})_2]$ cannot show geometrical isomerism.

Option (B) : $[\text{Zn}(\text{NH}_3)_2\text{Cl}_2]$ is a tetrahedral complex, cannot show geometrical isomerism.

Option (C) : $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ is an octahedral complex, can show geometrical isomerism.

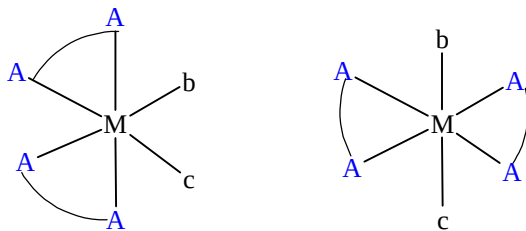


cis – form



trans – form

Option (D) : $[\text{Cr}(\text{en})_2(\text{H}_2\text{O})(\text{SO}_4)]^+$ is an octahedral complex and is of type $[\text{M}(\text{AA})_2\text{bc}]$, can show geometrical isomerism.



- Q.6 Atoms of metals x, y, and z form face-centred cubic (fcc) unit cell of edge length L_x , body-centred cubic (bcc) unit cell of edge length L_y , and simple cubic unit cell of edge length L_z , respectively.

$$\text{If } r_z = \frac{\sqrt{3}}{2} r_y; r_y = \frac{8}{\sqrt{3}} r_x; M_z = \frac{3}{2} M_y \text{ and } M_z = 3M_x,$$

then the correct statement(s) is(are)

[Given: M_x , M_y , and M_z are molar masses of metals x, y, and z, respectively.

r_x , r_y , and r_z are atomic radii of metals x, y, and z, respectively.]

- (A) Packing efficiency of unit cell of x > Packing efficiency of unit cell of y > Packing efficiency of unit cell of z
 (B) $L_y > L_z$
 (C) $L_x > L_y$
 (D) Density of x > Density of y

Sol. A, B, D

For metal 'x'

Fcc: Edge length, $a_1 = L_x$

For metal 'y'

Bcc: Edge length, $a_2 = L_y$

For metal 'z'

Bcc: Edge length, $a_3 = L_z$

$$r_z = \frac{\sqrt{3}}{2} r_y, r_y = \frac{8}{\sqrt{3}} r_x, M_z = \frac{3}{2} M_y \text{ and } M_z = 3M_x$$

For option (A)

- (i) For FCC ($Z = 4$) metal 'x', $4r_x = \sqrt{2}L_x$

$$\text{P.E} = \frac{Z \times \frac{4}{3} \pi (r_x)^3}{a_1^3} = \frac{4 \times \frac{4}{3} \pi (r_x)^3}{(L_x)^3} = \frac{4 \times \frac{4}{3} \pi (r_x)^3}{\left(\frac{4}{\sqrt{2}} r_x\right)^3} = 0.24\pi$$

- (ii) For BCC ($Z = 2$) metal 'y', $4r_y = \sqrt{3}L_y$

$$\text{P.E} = \frac{Z \times \frac{4}{3} \pi (r_y)^3}{a_2^3} = \frac{2 \times \frac{4}{3} \pi (r_y)^3}{(L_y)^3} = \frac{2 \times \frac{4}{3} \pi (r_y)^3}{\left(\frac{4}{\sqrt{3}} r_y\right)^3} = 0.22\pi$$

- (iii) For SC ($Z = 1$) metal 'z', $2r_z = L_z$

$$\text{P.E} = \frac{Z \times \frac{4}{3} \pi (r_z)^3}{a_3^3} = \frac{1 \times \frac{4}{3} \pi (r_z)^3}{(L_z)^3} = \frac{1 \times \frac{4}{3} \pi (r_z)^3}{(2r_z)^3} = \frac{\pi}{6} = 0.17\pi$$

$$(\text{P.E})_{\text{FCC}} > (\text{P.E})_{\text{BCC}} > (\text{P.E})_{\text{SC}}$$

For option (B)

$$4r_y = \sqrt{3}L_y \quad 2r_z = L_z$$

$$L_y = \frac{4r_y}{\sqrt{3}}$$

$$\frac{L_y}{L_z} = \frac{4r_y}{\sqrt{3} \times 2r_z} = \frac{2r_y}{\sqrt{3}r_z} = \frac{2r_y}{\sqrt{3} \cdot \frac{\sqrt{3}}{2} r_y} = \frac{4}{3}$$

So, $L_y > L_z$

For option (C)

$$4r_x = \sqrt{2}L_x, \quad 4r_y = \sqrt{3}L_y$$

$$L_x = \frac{4r_x}{\sqrt{2}}, \quad L_y = \frac{4r_y}{\sqrt{3}}$$

$$\frac{L_x}{L_y} = \frac{\sqrt{3}r_x}{\sqrt{2} \times 8 / \sqrt{3}r_x} = \frac{3}{8\sqrt{2}}$$

So, $L_x < L_y$ incorrect

For option (D)

$$d_x = \frac{4 \times M_x}{\left(\frac{4r_x}{\sqrt{2}}\right)^3 \times N_A}$$

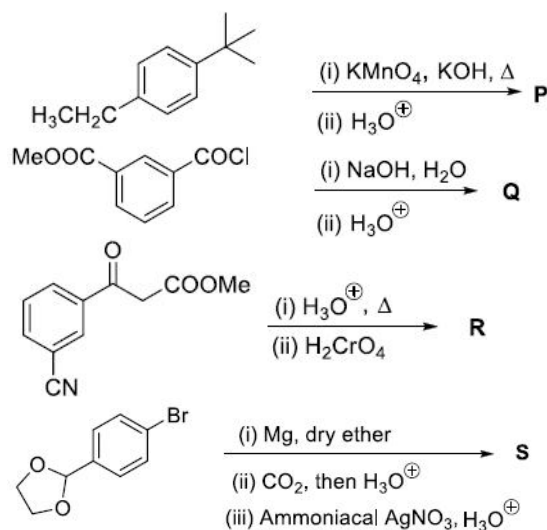
$$d_y = \frac{2 \times M_y}{\left(\frac{4r_y}{\sqrt{3}}\right)^3 \times N_A}$$

$$r_y = \frac{8}{\sqrt{3}}r_x, \quad \frac{M_x}{M_y} = \frac{1}{2}$$

$$\frac{d_x}{d_y} = \frac{512}{2\sqrt{2}} = \frac{256}{\sqrt{2}}$$

So $d_x > d_y$ (correct)

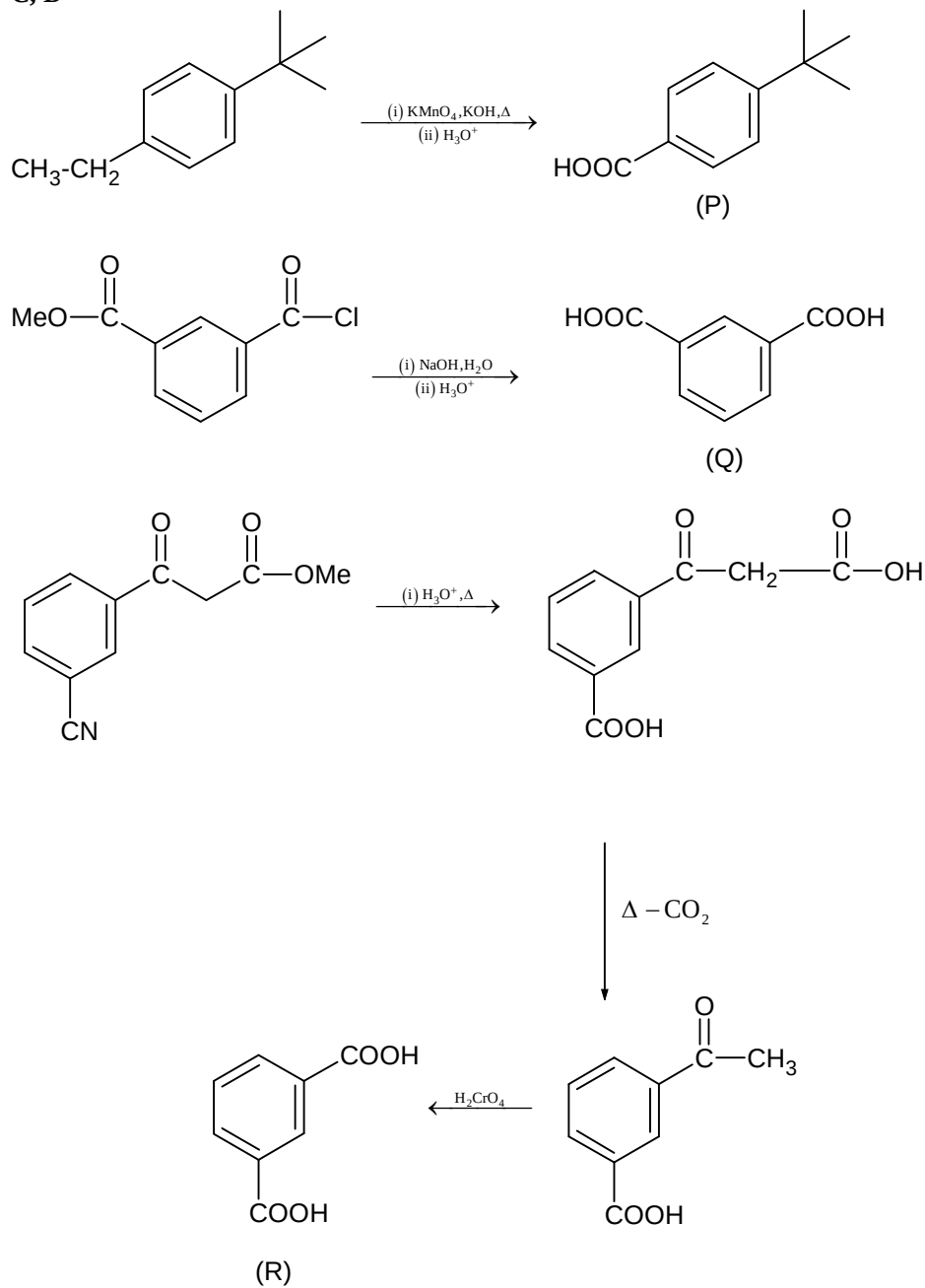
Q.7 In the following reactions, **P**, **Q**, **R**, and **S** are the major products.

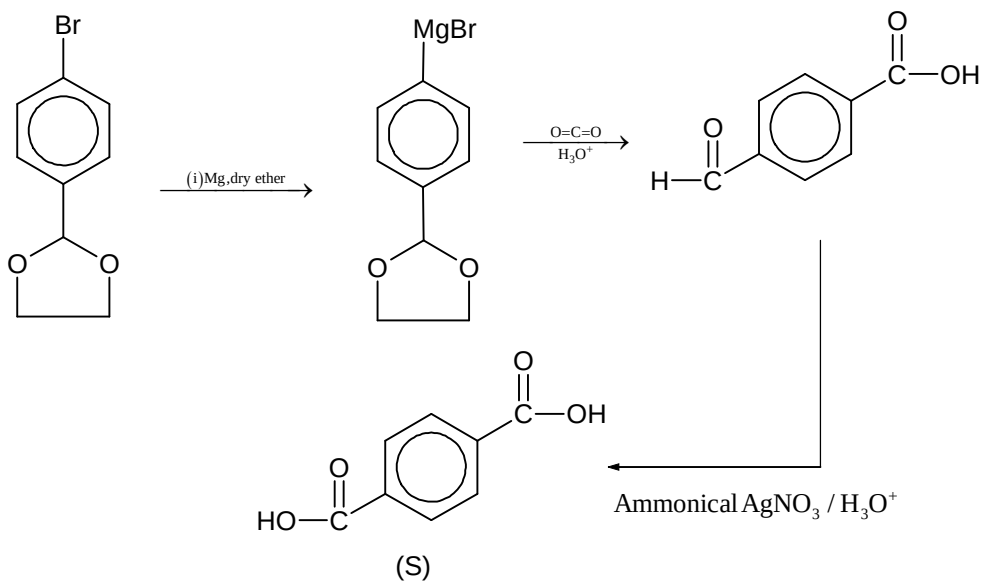


The correct statement(s) about **P**, **Q**, **R**, and **S** is(are)

- (A) **P** and **Q** are monomers of polymers dacron and glyptal, respectively.
 (B) **P**, **Q**, and **R** are dicarboxylic acids.
 (C) Compounds **Q** and **R** are the same.
 (D) **R** does **not** undergo aldol condensation and **S** does **not** undergo Cannizzaro reaction.

Sol. C, D

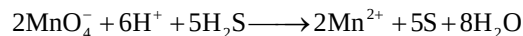
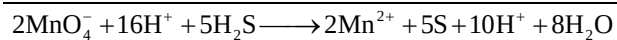
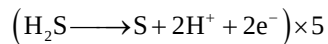
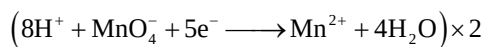


**SECTION 3 (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct integer is entered;
Zero Marks : 0 In all other cases.

*Q.8 H_2S (5 moles) reacts completely with acidified aqueous potassium permanganate solution. In this reaction, the number of moles of water produced is x , and the number of moles of electrons involved is y . The value of $(x + y)$ is ____.

Sol. 18

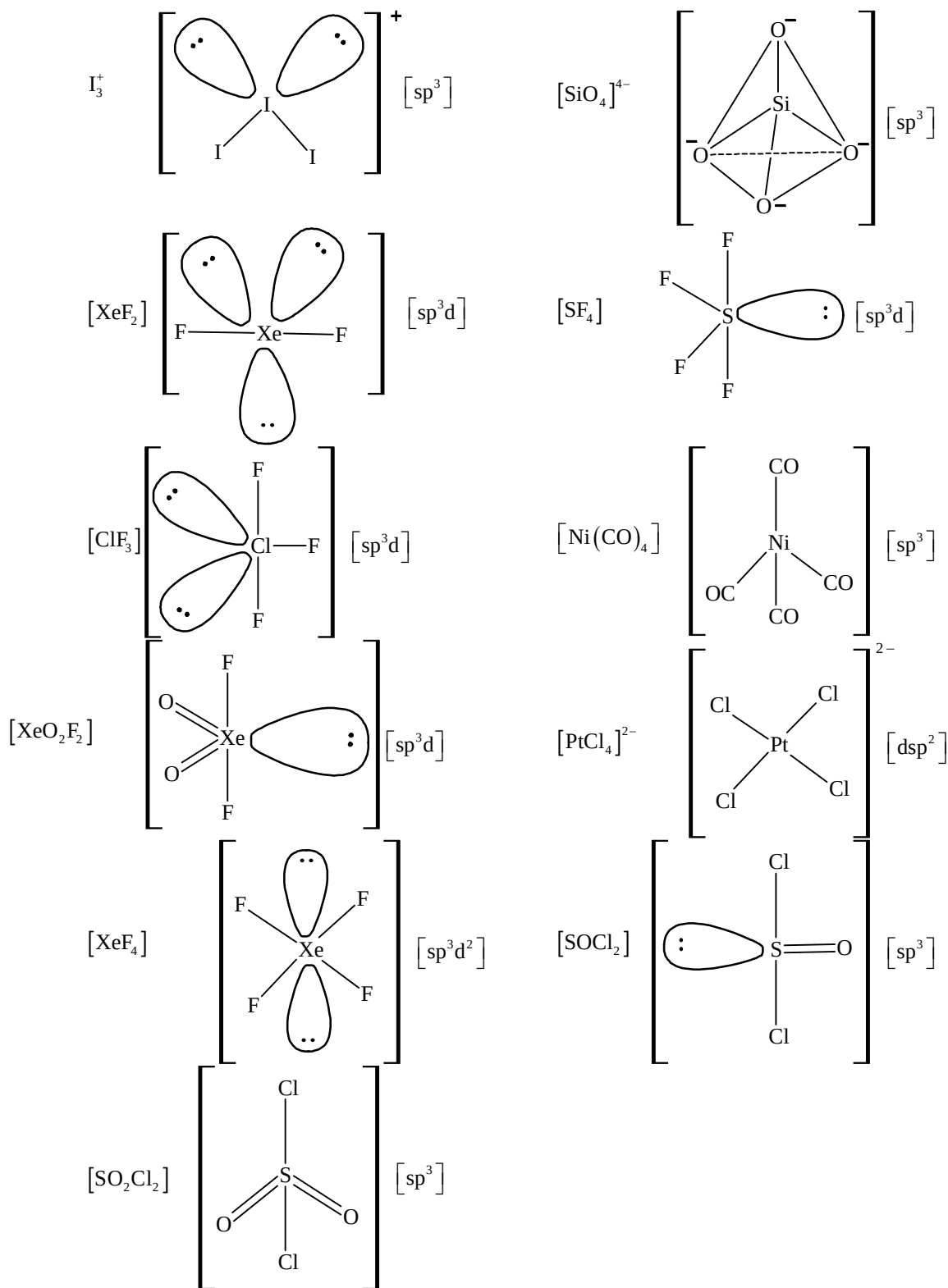


$$x = 8$$

$$y = 10$$

Q.9 Among $[\text{I}_3]^+$, $[\text{SiO}_4]^{4-}$, SO_2Cl_2 , XeF_2 , SF_4 , ClF_3 , $\text{Ni}(\text{CO})_4$, XeO_2F_2 , $[\text{PtCl}_4]^{2-}$, XeF_4 , and SOCl_2 , the total number of species having sp^3 hybridised central atom is ____.

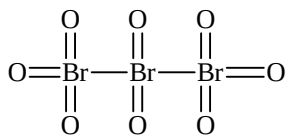
Sol. 5



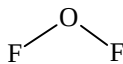
*Q.10 Consider the following molecules: Br_3O_8 , F_2O , $H_2S_4O_6$, $H_2S_5O_6$, and C_3O_2 .
Count the number of atoms existing in their zero oxidation state in each molecule.
Their sum is ____.

Sol. 06

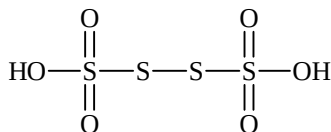
(i)



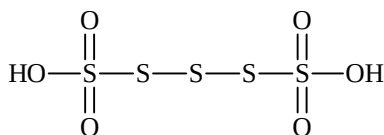
(ii)



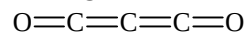
(iii)



(iv)



(v)



Total number of atoms having zero oxidation state = 6.

*Q.11 For He^+ , a transition takes place from the orbit of radius 105.8 pm to the orbit of radius 26.45 pm. The wavelength (in nm) of the emitted photon during the transition is ____.

[Use:

Bohr radius, $a = 52.9 \text{ pm}$ Rydberg constant, $R_H = 2.2 \times 10^{-18} \text{ J}$ Planck's constant, $h = 6.6 \times 10^{-34} \text{ J s}$ Speed of light, $c = 3 \times 10^8 \text{ m s}^{-1}$]**Sol. 30**

$$r_n = \frac{52.9 \times n^2}{Z} \text{ pm}$$

$$105.8 = \frac{52.9 \times n_1^2}{2} \quad \therefore n_1^2 = 4, n_1 = 2$$

$$26.45 = \frac{52.9 \times n_2^2}{2} \quad \therefore n_2 = 1$$

$$\frac{1}{\lambda} = 109677 \times 4 \times \frac{3}{4}$$

$$\lambda = \frac{4}{109677 \times 4 \times 3} \text{ cm}$$

$$= \frac{10^7}{109677 \times 3} = \frac{10^7}{329031} \text{ nm}$$

$$\lambda = 30.3 \text{ nm} \approx 30 \text{ nm}$$

Q.12 50 mL of 0.2 molal urea solution (density = 1.012 g mL^{-1} at 300 K) is mixed with 250 mL of a solution containing 0.06 g of urea. Both the solutions were prepared in the same solvent. The osmotic pressure (in Torr) of the resulting solution at 300 K is ____.

[Use: Molar mass of urea = 60 g mol^{-1} ; gas constant, $R = 62 \text{ L Torr K}^{-1} \text{ mol}^{-1}$;Assume, $\Delta_{\text{mix}} H = 0$, $\Delta_{\text{mix}} V = 0$]**Sol. 682**

0.2 molal means 0.2 moles in 1000 g of solvent.

$$\text{Volume} = \frac{M}{d}$$

$$\text{Mass of solution} = 1012 \text{ g}$$

$$\text{Volume} = \frac{1012 \text{ g}}{1.012 \text{ g ml}^{-1}}$$

$$V = 1000.00 \text{ ml}$$

$$1000.00 \text{ ml} \longrightarrow 0.2 \text{ moles}$$

$$50 \text{ ml of solution} = \frac{0.2}{1000} \times 50 \text{ moles}$$

$$n_{\text{urea}} = 0.01 \text{ moles}$$

In 2nd solution:

$$n_{\text{urea}} = \frac{0.06}{60} = 0.001$$

$$\text{Final molarity (M)} = \frac{n_1 + n_2}{V_1 + V_2} = \frac{0.01 + 0.001}{\frac{(50 + 250)}{1000}}$$

$$M = \frac{11}{300}$$

$$\pi = \text{CRT}$$

$$= \frac{11}{300} \times 62 \times 300$$

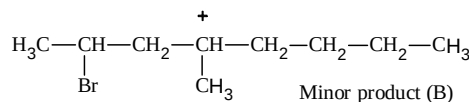
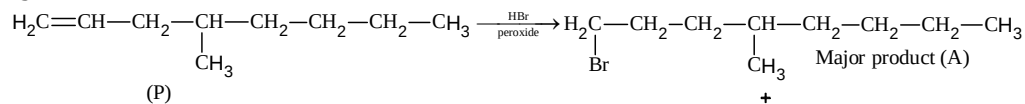
$$= 682 \text{ torr}$$

- Q.13 The reaction of 4-methyloct-1-ene (**P**, 2.52 g) with HBr in the presence of $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$ gives two isomeric bromides in a 9 : 1 ratio, with a combined yield of 50%. Of these, the entire amount of the primary alkyl bromide was reacted with an appropriate amount of diethylamine followed by treatment with aq. K_2CO_3 to give a non-ionic product **S** in 100% yield.

The mass (in mg) of **S** obtained is ____.

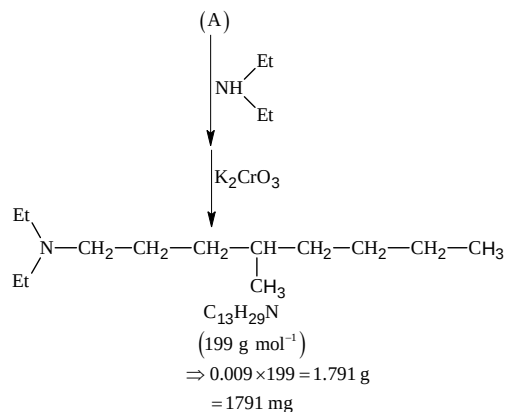
[Use molar mass (in g mol^{-1}): H = 1, C = 12, N = 14, Br = 80]

Sol. 1791



$$\text{Moles of P} = \frac{2.52}{126} = 0.02 \text{ mole}$$

$$\text{Moles of A} = 0.02 \times \frac{9}{10} \times \frac{50}{100} = 0.009 \text{ mole}$$

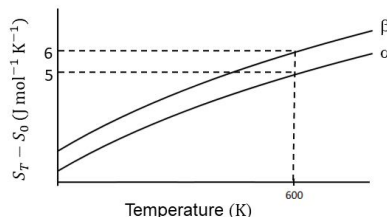


SECTION 4 (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, there are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered in the designated place;
Zero Marks : 0 In all other cases.

"PARAGRAPH I"

The entropy versus temperature plot for phases α and β at 1 bar pressure is given.
 S_T and S_0 are entropies of the phases at temperatures T and 0 K, respectively.



The transition temperature for α to β phase change is 600 K and $C_{p,\beta} - C_{p,\alpha} = 1 \text{ J mol}^{-1} \text{ K}^{-1}$. Assume $(C_{p,\beta} - C_{p,\alpha})$ is independent of temperature in the range of 200 to 700 K. $C_{p,\alpha}$ and $C_{p,\beta}$ are heat capacities of α and β phases, respectively.

- *Q.14 The value of entropy change, $S_\beta - S_\alpha$ (in $\text{J mol}^{-1} \text{ K}^{-1}$), at 300 K is ____.
 [Use: $\ln 2 = 0.69$
 Given: $S_\beta - S_\alpha = 0$ at 0 K]

Sol. 0.31

$$S = S_0 + \int C_p \frac{dT}{T}$$

$$S_\alpha = S_0 + \int (C_p)_\alpha \frac{dT}{T}$$

$$S_\beta = S_0 + \int (C_p)_\beta \frac{dT}{T}$$

$$S_\beta - S_\alpha = \left[(C_p)_\beta - (C_p)_\alpha \right] \int \frac{dT}{T}$$

$$\text{Given } C_{p,\beta} - C_{p,\alpha} = 1$$

$$S_\beta - S_\alpha = \ln T + C \text{ at any temperature } T.$$

$$(S_\beta - S_\alpha)_{T_2} - (S_\beta - S_\alpha)_{T_1} = \ln T_2 - \ln T_1$$

$$T_2 = 600 \text{ K}, T_1 = 300 \text{ K, from the graph } S_\beta - S_\alpha \text{ at } 600^\circ\text{C} = 1$$

$$(1) - (S_\beta - S_\alpha)_{300} = \ln 600 - \ln 300$$

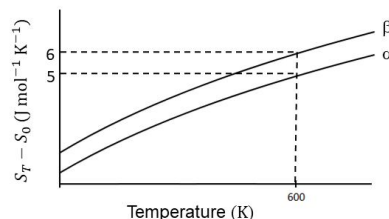
$$1 - (S_\beta - S_\alpha)_{300} = \ln 2 = 0.69$$

$$\Rightarrow (S_\beta - S_\alpha)_{300} = 1 - 0.69$$

$$= 0.31$$

"PARAGRAPH I"

The entropy versus temperature plot for phases α and β at 1 bar pressure is given.
 S_T and S_0 are entropies of the phases at temperatures T and 0 K, respectively.



The transition temperature for α to β phase change is 600 K and $C_{p,\beta} - C_{p,\alpha} = 1 \text{ J mol}^{-1} \text{ K}^{-1}$. Assume $(C_{p,\beta} - C_{p,\alpha})$ is independent of temperature in the range of 200 to 700 K. $C_{p,\alpha}$ and $C_{p,\beta}$ are heat capacities of α and β phases, respectively.

*Q.15 The value of enthalpy change, $H_\beta - H_\alpha$ (in J mol^{-1}), at 300 K is ____.

Sol. 300

Transition: $\alpha \rightleftharpoons \beta$; $\Delta G = 0$

So, $\Delta H = T\Delta S$

$$\Delta H_{600} = 600 \times 1 \quad \because \Delta S = 1$$

$$= 600 \text{ J mol}^{-1}$$

From Krichoff's law

$$\Delta C_p = \frac{\Delta H_{600} - \Delta H_{300}}{600 - 300}$$

$$1 = \frac{600 - \Delta H_{300}}{300}$$

$$\Delta H_{300} = 300 \text{ J mol}^{-1}$$

"PARAGRAPH II"

A trinitro compound, 1,3,5-tris-(4-nitrophenyl)benzene, on complete reaction with an excess of Sn/HCl gives a major product, which on treatment with an excess of NaNO_2/HCl at 0°C provides **P** as the product. **P**, upon treatment with excess of H_2O at room temperature, gives the product **Q**. Bromination of **Q** in aqueous medium furnishes the product **R**. The compound **P** upon treatment with an excess of phenol under basic conditions gives the product **S**.

The molar mass difference between compounds **Q** and **R** is 474 g mol^{-1} and between compounds **P** and **S** is 172.5 g mol^{-1} .

Q.16 The number of heteroatoms present in one molecule of **R** is ____.

[Use: Molar mass (in g mol^{-1}): H = 1, C = 12, N = 14, O = 16, Br = 80, Cl = 35.5
 Atoms other than C and H are considered as heteroatoms]

Sol. 9

“PARAGRAPH II”

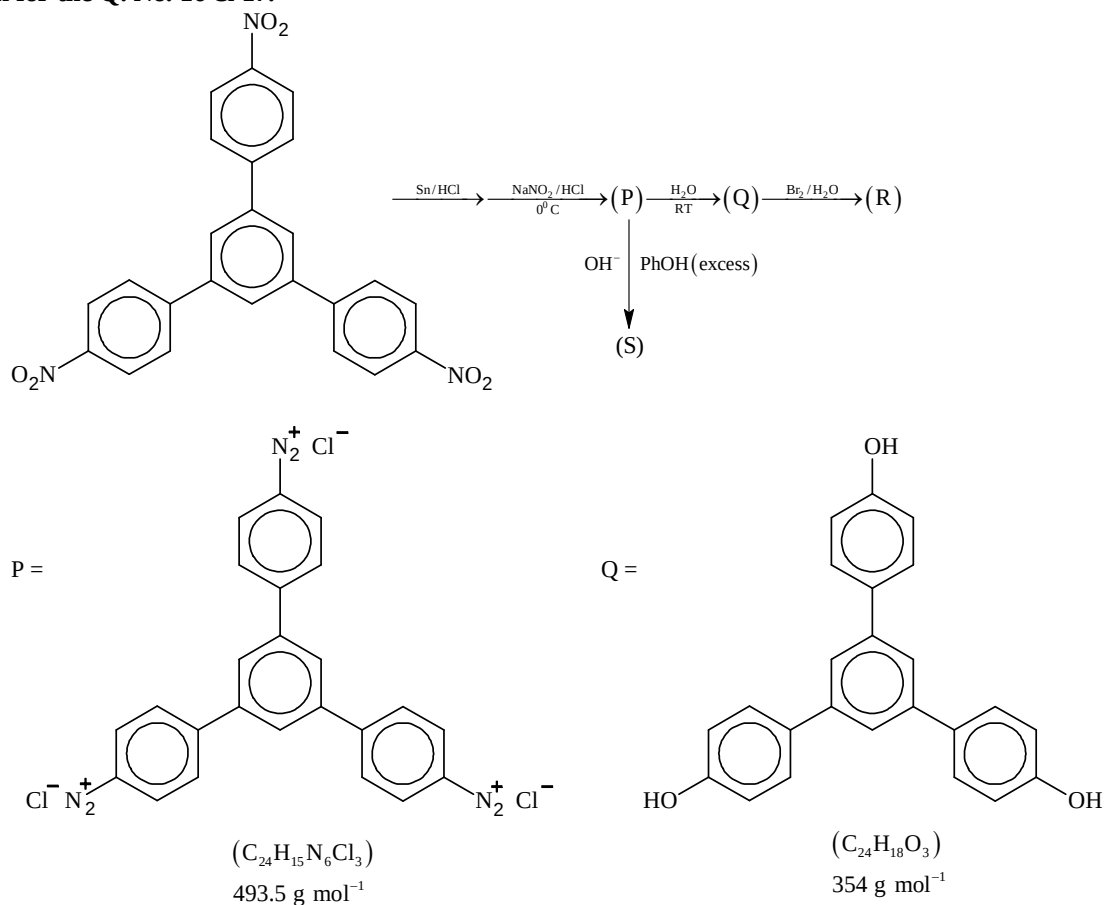
A trinitro compound, 1,3,5-tris-(4-nitrophenyl)benzene, on complete reaction with an excess of Sn/HCl gives a major product, which on treatment with an excess of NaNO₂/HCl at 0°C provides **P** as the product. **P**, upon treatment with excess of H₂O at room temperature, gives the product **Q**. Bromination of **Q** in aqueous medium furnishes the product **R**. The compound **P** upon treatment with an excess of phenol under basic conditions gives the product **S**.

The molar mass difference between compounds **Q** and **R** is 474 g mol⁻¹ and between compounds **P** and **S** is 172.5 g mol⁻¹.

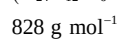
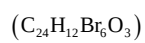
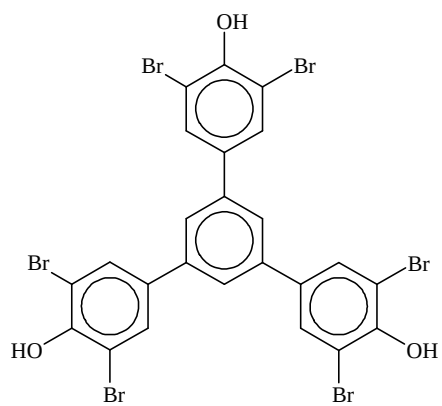
- Q.17 The total number of carbon atoms and heteroatoms present in one molecule of **S** is _____.
 [Use: Molar mass (in g mol⁻¹): H = 1, C = 12, N = 14, O = 16, Br = 80, Cl = 35.5
 Atoms other than C and H are considered as heteroatoms]

Sol. 51

Solution for the Q. No. 16 & 17.



R =



S =

